

Yu-Lin Liu

willy229.liu@gmail.com | Madison, WI (Fall 2026) / Kaohsiung, Taiwan | linkedin.com/in/willy229liu | willy229liu.github.io

EDUCATION

University of Wisconsin-Madison | Madison, WI

Master of Science (M.S.) in Computer Science | Sep 2026 - May 2028 (Expected)

Nanjing University | Nanjing, China

Bachelor of Engineering (B.E.) in Software Engineering | Sep 2018 - Jun 2022 | GPA: 4.16/5.0

- Coursework: Data Structures & Algorithms, Computer Architecture, Operating System, Machine Learning, Pattern Recognition, Software Engineering, Software System Design, HCI, C++ Programming Language

EXPERIENCE

Lilith Games | Game Engine Development Engineer, AFK Journey | Jul 2022 - Apr 2025

- Built C++, C#, Lua, and Python runtime, performance, and developer tools for AFK Journey, an award-winning open-world mobile game used by millions of players.
- Cut runtime memory 60% (2.5 GB to 1.0 GB) by profiling allocations, replacing high-overhead containers, adopting memory-mapped loading, and redesigning data pipelines.
- Extended stable sessions on memory-constrained 3 GB devices beyond one hour by identifying retained objects and memory leaks, reducing allocation churn, and mitigating fragmentation.
- Cut startup latency 75% (40s to 10s) by redesigning data parsing, asynchronous I/O, and resource-loading workflows across startup-critical pipelines.
- Cut animation runtime memory 85% and per-asset size 90% through demand loading, lifecycle management, and compression, reducing persistent memory residency.
- Optimized a multithreaded small-map renderer for 500+ dynamic objects using LOD, vertex reduction, camera smoothing, and draw-call reduction to maintain responsive interaction.
- Built automated asset-processing, FlatBuffers export, resource-management, and visual-effect performance-grading tools supporting production content workflows.

Lilith Games | Game Engine Development Intern, AFK Journey | Jul 2021 - Feb 2022

- Built Infinite Map Zoom with responsive boundary drag and pinch, seamless 3D-to-2D transitions, and camera, map, and icon gesture logic without noticeable loading.
- Implemented rule-based 2D mini-map icon rendering and built map-copy, FlatBuffers export, and resource-management tools supporting designer and artist workflows.

SELECTED PROJECTS & RESEARCH

ECG Modeling Research

- Built a reproducible preprocessing and model-comparison pipeline for noisy single-lead ECG from hospital PSG studies; evaluated ResNet-18, MultiRocket, TabPFN, XGBoost, and foundation models to estimate AHI, an indicator of sleep apnea severity.

UI Fault Detection Capstone

- Generated synthetic rendering faults from real mobile application screenshots and trained a 2D CNN classifier to detect UI rendering defects, achieving 0.87 test accuracy.

Huawei CodeCraft Algorithm Competition | National Third Prize

- Improved the time complexity of C++ directed-graph cycle detection and centrality-ranking algorithms.

TECHNICAL SKILLS

Frameworks & Platforms: Unity, PyTorch, XGBoost, Windows, Android, iOS

Languages: C++, Python, C#, Lua, Java

Systems & Performance: multithreading, memory profiling, CPU profiling, memory-mapped I/O, asynchronous I/O, cache-aware data structures, serialization, benchmarking, virtual machine optimization

Developer Tools: FlatBuffers, Git, Perforce, Jenkins